

CUSTOMER SEGMENTATION USING K-MEANS & CRB

MBSA PROJECT

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1. Source of Data and Its Range:

The data consists of 200 customers with information related to their age, gender, annual income, and spending score. The spending score is a numeric variable ranging from 1 to 100 and was assigned to customers based on behavior parameters and purchasing data. The data set also contains the customer's ID, which will be dropped before beginning the analysis.

Source: Typically, this dataset is publicly available from machine learning repositories like Kaggle. It includes:

- CustomerID: Unique ID for each customer.
- Gender: Male/Female.
- Age: Ranges from 18 to 70 years.
- Annual Income (k\$): Ranges from 15k\$ to 137k\$.
- Spending Score (1-100): A value assigned by the mall based on customer behavior and spending.

2. Basic EDA (Exploratory Data Analysis):

The dataset contains 200 customers with features: Age, Gender, Annual Income (k\$), and Spending Score (1–100). All data is clean and complete.

- **Age:** Mostly between 20 and 40 years.
- **Annual Income:** Ranges from 15k\$ to 137k\$, showing two income groups.
- **Spending Score:** Ranges from 1 to 99, also shows two major spending patterns.

- **Gender:** Almost equal number of males and females.

There is no strong correlation between income and spending, which supports using clustering to group customers with similar behavior.

Key Observations:

- High spenders are not necessarily high-income individuals.
- There is no direct correlation between income and spending, indicating complex customer behavior suitable for clustering.
- Scatter plots and 3D plots show visually separable clusters.

3. Signal and Probabilistic Model:

The data can be thought of as signals representing customer behavior across measurable variables like age, income, and spending. Each customer is treated as a data point in a multidimensional space.

A probabilistic approach assumes that these data points are generated from a mixture of distributions (e.g., clusters), where each distribution represents a different type of customer segment. K-Means assumes these clusters are spherical and well-separated.

4. Model Assumptions:

K-Means clustering makes several key assumptions:

- Clusters are spherical and equally sized.
- Each feature contributes equally to distance calculations (after standardization).
- The number of clusters (k) is pre-specified, typically selected using the Elbow Method.
- There are no significant outliers skewing the cluster centroids.

These assumptions work reasonably well for the current dataset, especially after feature scaling.

5. Common Models Used For The Data Under Consideration:

In this project, **K-Means Clustering** is the chosen model for customer segmentation. It is a widely used unsupervised learning algorithm that groups similar data points into **k distinct**

clusters based on feature similarity. In our case, the features considered are **Age**, **Annual Income (k\$)**, and **Spending Score (1–100)**.

Why K-Means?

- **Simplicity and Efficiency:** K-Means is computationally efficient and easy to implement, making it ideal for large datasets.
- **Interpretability:** The resulting clusters provide clear groupings of customer types based on behavior and demographics.
- **Speed:** It converges quickly compared to many other clustering algorithms.

How It Works (Conceptually):

1. The algorithm selects **k initial centroids**.
2. Each data point is assigned to the nearest centroid, forming k clusters.
3. New centroids are calculated as the mean of the points in each cluster.
4. Steps 2 and 3 are repeated until the centroids no longer change significantly (i.e., the algorithm converges).

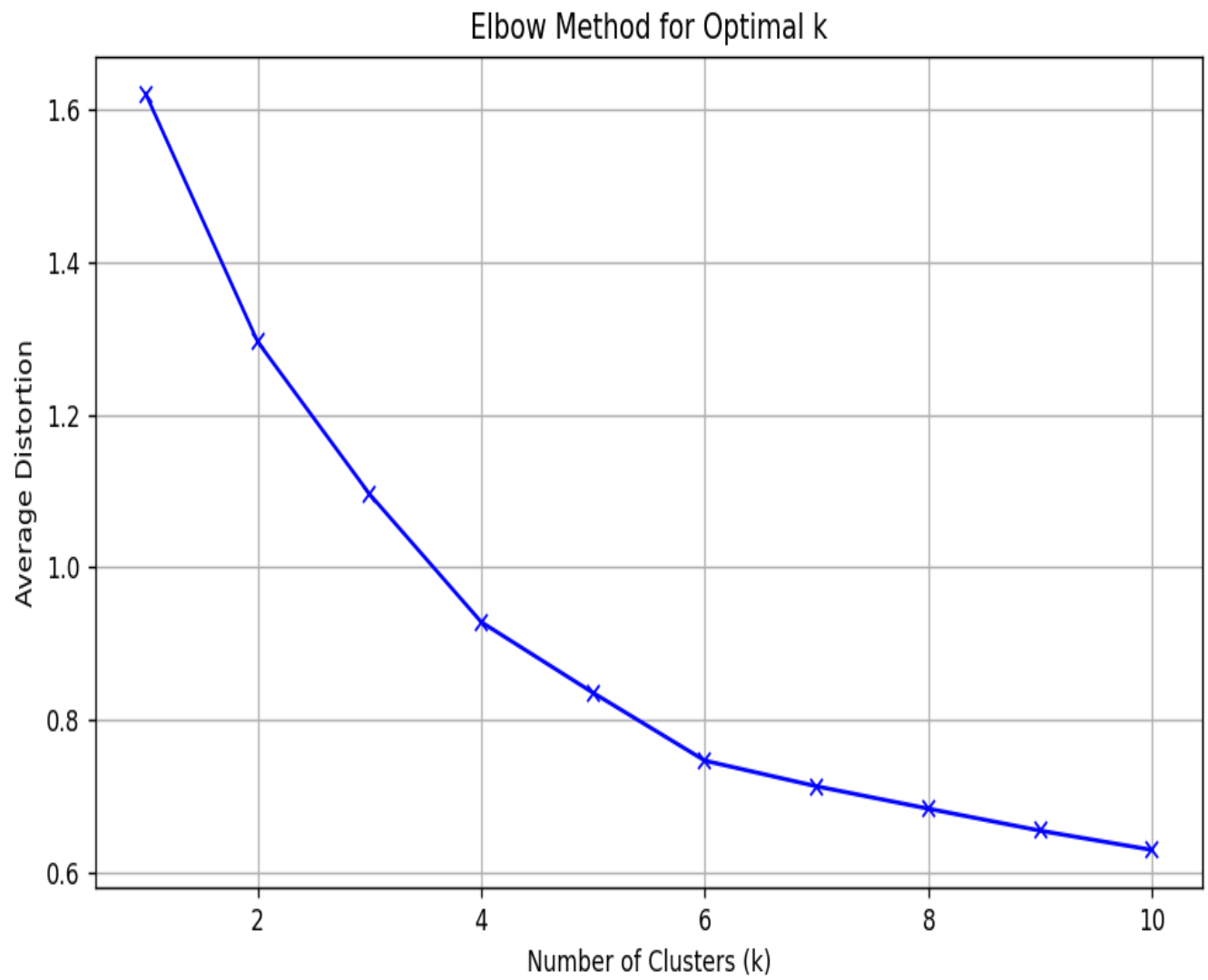
Application in This Project:

- The **Elbow Method** was used to determine the optimal number of clusters (k).
- A **3D visualization** was generated using standardized values to observe the natural grouping of customers.
- An additional **bar graph** shows the distribution of customers across the clusters, helping interpret the segmentation results.

Insights from Using K-Means:

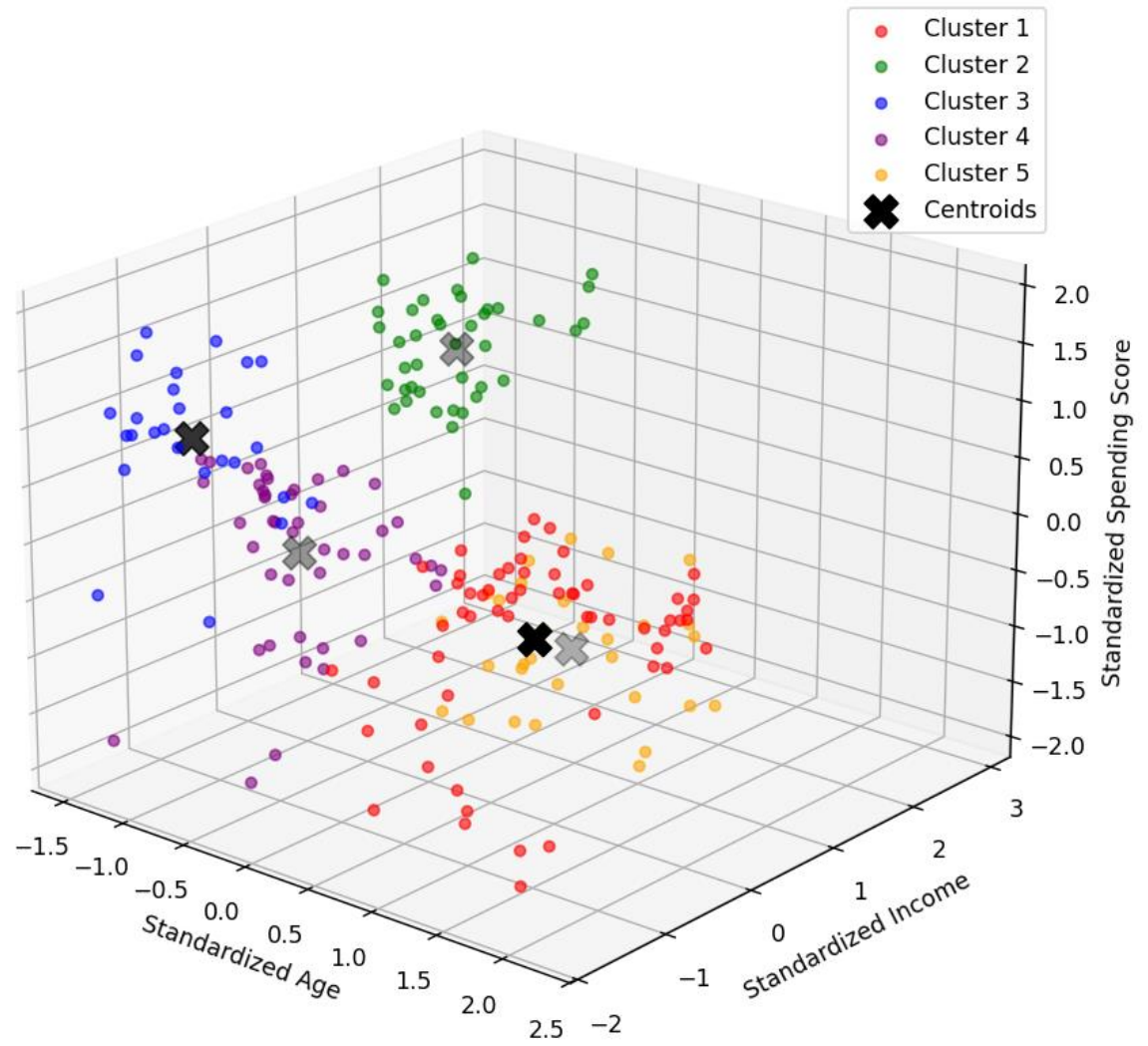
- The model revealed natural groupings of customers based on their age, income, and spending habits.
- Businesses can target each cluster with tailored marketing strategies (e.g., high spenders vs. budget-conscious customers).

OUTPUTS:



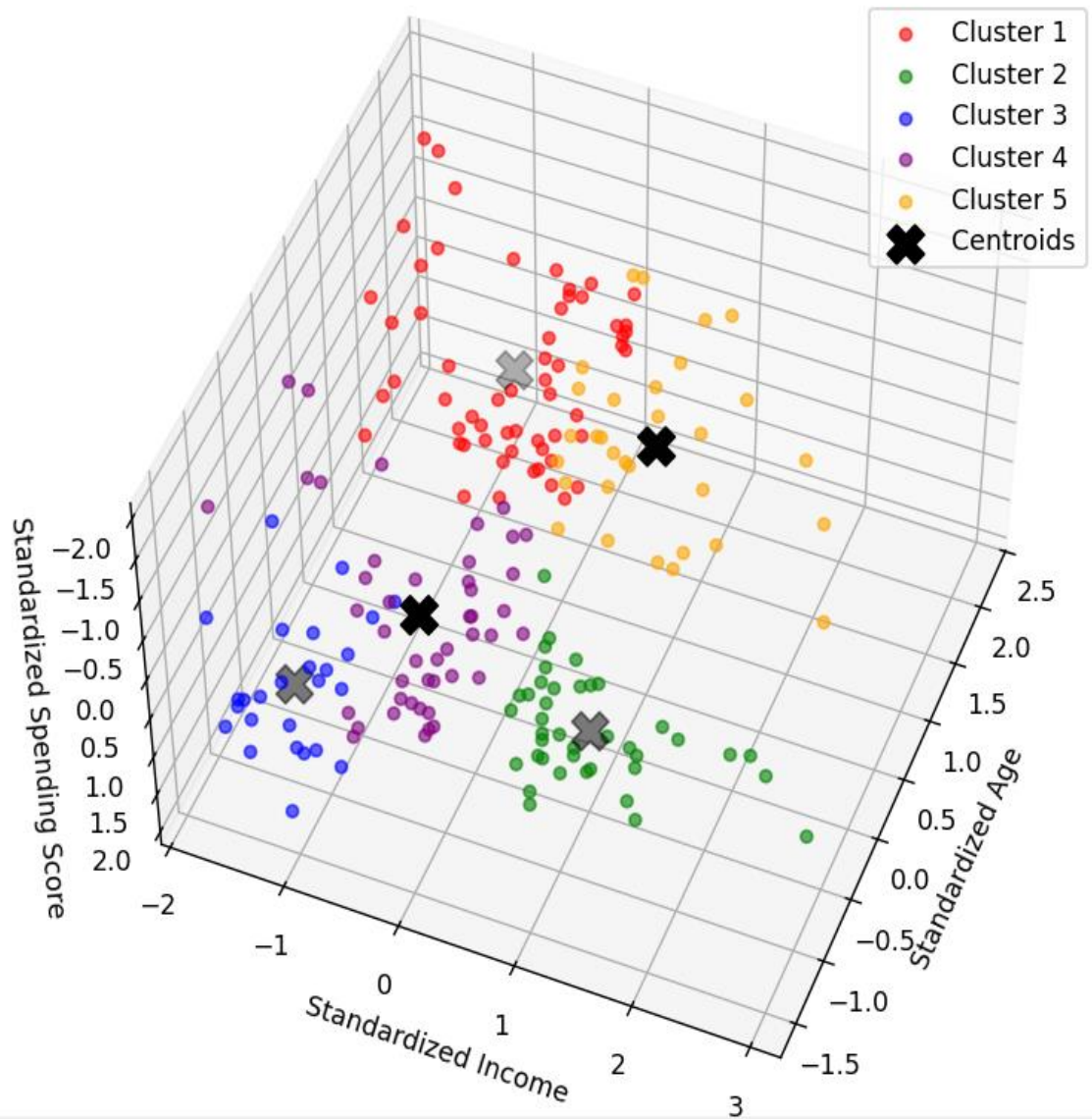
ONE SIDE VIEW

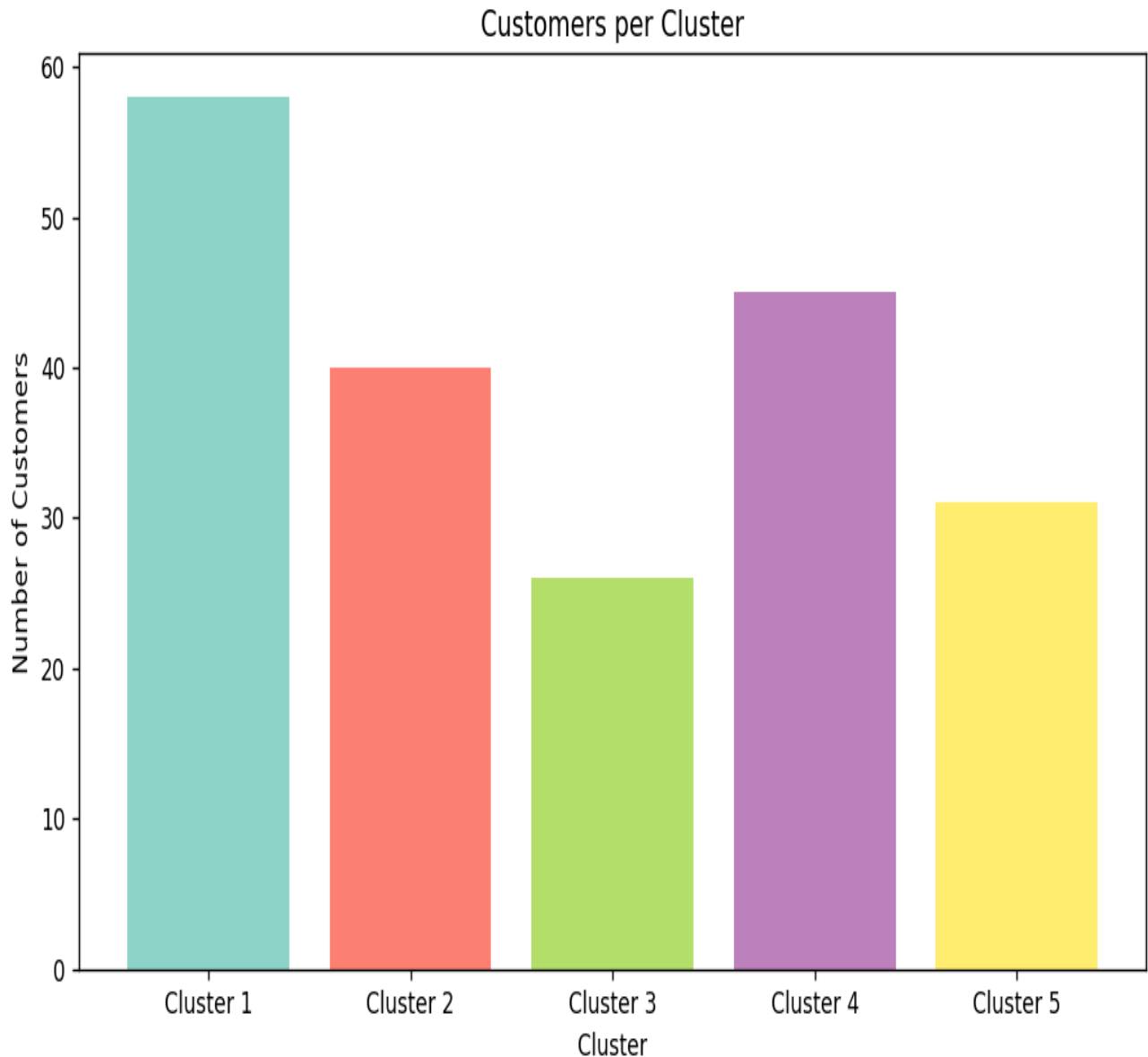
Customer Segments (3D View)



OTHER SIDE VIEW

Customer Segments (3D View)





Final Conclusion:

The customer segmentation project using **K-Means Clustering** successfully grouped mall customers into five distinct clusters based on **Age**, **Annual Income**, and **Spending Score**. The optimal number of clusters ($k = 5$) was identified using the **Elbow Method**, and the 3D visualization clearly demonstrated meaningful separation among customer groups. Additionally, a bar graph illustrated the distribution of customers across these clusters, making it easier to interpret customer density per segment.

These insights can be instrumental for businesses in tailoring marketing strategies, offering personalized services, and enhancing customer satisfaction. For example:

- One cluster may represent **young high spenders**, ideal for premium promotions.
- Another cluster might consist of **older low spenders**, suitable for cost-saving offers.

Overall, the project provides a strong foundation for data-driven decision-making in customer relationship management.